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Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *arXiv preprint arXiv:2306.05685*.

Hao Zhou, Vamsi K Ithapu, Sathya Narayanan Ravi, Vikas Singh, Grace Wahba, and Sterling C Johnson. 2016. [Hypothesis testing in unsupervised domain adaptation with applications in alzheimer's disease](#). In *Advances in Neural Information Processing Systems*, volume 29. Curran Associates, Inc.

Andy Zou, Tristan Xiao, Ryan Jia, Joe Kwon, Mantas Mazeika, Richard Li, Dawn Song, Jacob Steinhardt, Owain Evans, and Dan Hendrycks. 2022. [Forecasting future world events with neural networks](#).

A Related Work

Given the limitations of traditional methods, researchers are exploring new ways to measure model performance more comprehensively. Recently, several methods have been proposed to evaluate models at different times and in various contexts [Fatemi et al. \(2024\)](#); [Kasai et al. \(2024\)](#); [Vu et al. \(2023\)](#). TimeQA [Chen et al. \(2021\)](#) evaluates a model’s ability to handle temporal evolution, while Situated QA [Zhang and Choi \(2021\)](#) focuses on a model’s performance in specific contexts. StreamingQA [Liška et al. \(2022\)](#) introduces the concept of continuous information flow, while RealTimeQA [Kasai et al. \(2024\)](#) and FreshQA [Vu et al. \(2023\)](#) emphasize real-time and up-to-date information processing capabilities. LatestEval [Li et al. \(2023b\)](#) utilizes the most recent text to construct questions and answers, covering six most frequently queried categories including explanation, summary, reason, demonstrations, existence, and possible usage.

However, it’s essential to clarify that RealTimeQA [Kasai et al. \(2024\)](#) and FreshQA [Vu et al. \(2023\)](#), which continuously update topics, are inclined towards knowledge. For example, RealTimeQA [Kasai et al. \(2024\)](#) has such a sample in their paper, "Which wildly popular show was recently greenlit for a new season?" (Answer: Squid Game). These methods are typically used for evaluating retrieval-augmented generation (rag) series models. On the other hand, methods like LatestEval [Li et al. \(2023b\)](#), which construct questions using the most recent information, lean towards enhancing reading comprehension.

In contrast to these approaches, our focus is on testing the model’s understanding of the world, as demonstrated through prediction. Our method of evaluation is based on facts and computation loss. This ensures that our evaluation is resistant to data contamination, objective, and free from biases in question construction and judgment. This approach allows us to examine the model’s ability to process and predict real-world events accurately and efficiently.

Future-knowledge Prediction Previous research has focused on how to automatically predict real-world events. For instance, [Zou et al. \(2022\)](#) constructed a dataset of questions obtained from tournament predictions as an evaluation benchmark for automatic forecasting. However,

this dataset may already be outdated because its knowledge is likely already embedded within current LLMs. Recently, studies have shown that LLMs can match human performance in the task of forecasting the future. For example, [Schoenegger et al. \(2024\)](#) demonstrates that the predictive performance of an ensemble of multiple LLMs is superior to that of a single LLM, and it even surpasses the performance of a group of human forecasters.

Model selection and Experimental Setting We selected a set of models due to their diversity and significance in advancing the capabilities of NLP systems. They are not only benchmarks of their respective sizes, ranging from 1 billion to 40 billion parameters, but also exemplify the rapid progression in NLP technology. We set the context size to 2048, then compute the sum of the negative logarithms of each segment. We then divide the computed sum by the length of the text under UTF-8 encoding. Throughout this experiment, approximately 300 GPU hours on NVIDIA A800 were consumed.

B Exploring Mitigation Strategies

While our primary focus is temporal evaluation rather than solution development, we conducted preliminary investigations into prompt engineering strategies that could potentially mitigate temporal performance degradation.

B.1 Try Different Prompt to Mitigate

Motivated by observed safety constraints and refusal patterns in base models and the result of [Pham and Cunningham \(2024\)](#), we designed five specialized prompts to investigate how different framing strategies affect temporal prediction:

- **Base:** Used direct incentivization ("tip of \$200") and explicit guessing encouragement to counter safety-driven refusals.
- **Dream Framework (v1):** Encourages speculative thinking by positioning the model as a visionary oracle.
- **Time Travel (v2):** Frames predictions as historical memories from a future perspective.
- **Multiverse (v3):** Leverages parallel universe observations to bypass temporal constraints.
- **Precise Thinking (v4):** Emphasizes structured temporal reasoning chains.
- **Future Recall (v5):** Simulates retrospective analysis from a future vantage point.

See Figure 6, 7, 8 and 9. Specifically, different models respond variably to the designed prompts. For Yi-6B, there were scarcely any discernible differences in performance across the several prompts we employed. In contrast, for the Qwen2.5-7B model, all modified prompts generally outperformed the base prompt in most scenarios. Analyzing the results from Llama3.1-8B-Instruct, Qwen2.5-7B and Baichuan2-7B-Chat, we found that prompts v5(Future Recall) and v4(Precise Thinking) demonstrated relatively better effectiveness. The specific impacts vary from model to model, which is closely related to the model's instruction-following capabilities.

In addition, our exploratory analysis revealed significant performance variations across question categories, with the model achieving lowest accuracy on "Finance" (58.1%), "Business" (55.4%), and "Economic Indicators" (62.3%) categories, compared to stronger performance on "Foreign Policy", (81.2%), "Security and Conflict" (79.6%) and "Leader Entry/Exit" (83.7%) categories.

It is plausible that fine-tuning models with analysis and prediction data (especially in these weak areas) could fully unlock their inner predictive potential.

C Full Table of Decline in Prediction Accuracy

See Table 4.

D Experimental setting

For this experiment, we sourced all questions from [Good Judgment Open](#) using web scraping. The data collected includes initiation and conclusion dates, domains, questions, descriptions, and answers, which were appropriately ordered for further processing. Approximately 50 GPU hours on NVIDIA A800 were consumed during this experiment.

Up to 2024.10.1, we collected a total of 2769 questions from [Good Judgment Open](#). The earliest question was proposed on 2015-09-01: "Will Congress pass a resolution disapproving the Joint Comprehensive Plan of Action?" The most recent question was closed on 2024-10-01: "What will be the 12-month percentage change in the US Consumer Price Index (CPI) for September 2024?" Question samples and most common tags are in Appendix J.

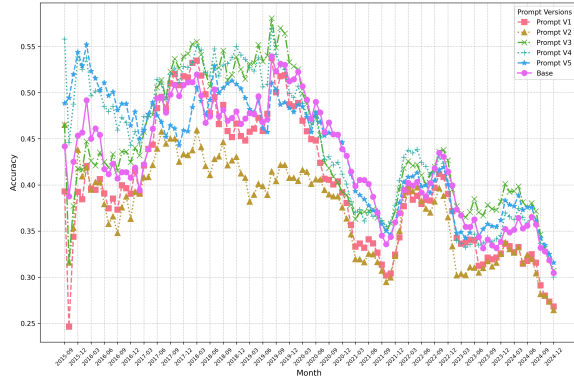


Figure 6: Accuracy of LLaMA3.1-8B-Instruct with different prompts

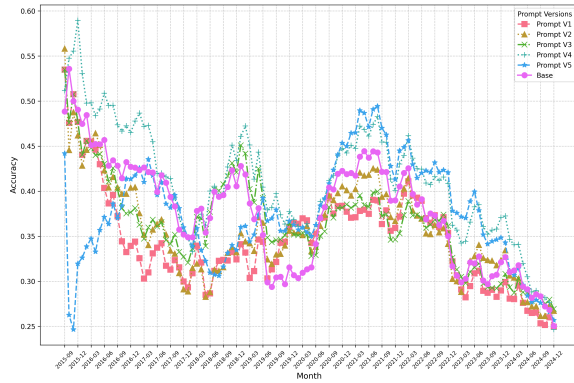


Figure 8: Accuracy of Baichuan2-7B-Chat with different prompts

E Testing Statistical Hypothesis

Due to the inherent randomness in data, directly comparing outcomes of two experiments is insufficient to justify the superiority of one outcome over the other. To address this issue, practitioners often resort to hypothesis testing to make convincing decisions, like those demonstrated in [Zhou et al. \(2016\)](#). A board application of hypothesis testing lies in the A/B testing literature, which assesses the performance of various features of a product and performs post-experiment inferences, as introduced in [Russac et al. \(2021\)](#).

F Details of Data Collection

The data for our system is sourced from a diverse set of online platforms, each selected for its relevance to different data types and domains. We utilize a Python-based crawling framework, which is detailed in our publicly available repository. This framework is designed to be adaptable, allowing for the inclusion of additional sources as they become relevant.

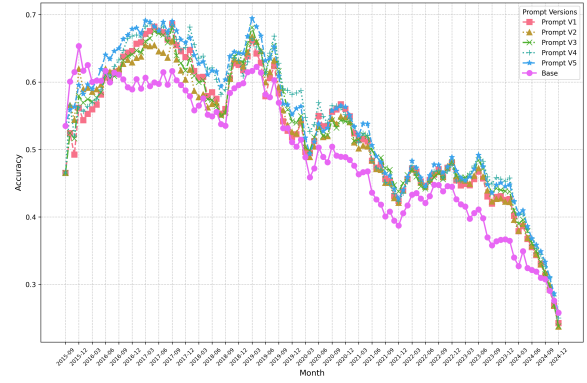


Figure 7: Accuracy of Qwen2.5-7B with different prompts

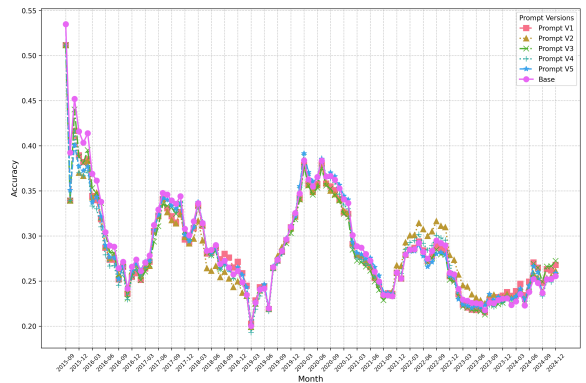


Figure 9: Accuracy of Yi-6B with different prompts

The primary data sources include:

- **Financial News:** Crawled from Yahoo, providing up-to-date information on global financial trends.
- **Political News:** BBC News is used to gather the latest political developments worldwide.
- **Discussion Forums:** Reddit is scraped to capture current discussions across a variety of tofigures.
- **Online Literature:** Wattpad offers a rich source of contemporary fiction and non-fiction.
- **Encyclopedia:** Wikipedia’s latest changes feed is monitored for updates across all domains.
- **Academic Papers:** arXiv is used to access the newest research across multiple scientific fields.
- **Code Repositories:** GitHub Trending provides insights into the latest developments in software engineering.